

The Association Between Healthcare Workers' Burnout and Biophysical Parameters

Interim Study Report — July 2026

MREC #3190 | SQU-EC297-2023 | Royal Hospital EC Approved PI: Dr. Mohamed Al Rawahi | Co-PI: Dr. Abdullah M. Al Alawi Sites: SQUH, Royal Hospital, Armed Forces Hospital — Muscat, Oman

1. Study Overview

A multi-center prospective pilot cohort study using the Copenhagen Burnout Inventory (CBI), validated psychological instruments (PHQ-9, GAD-7, ISI, WHO-5), and WHOOP wearable biosensors to quantify the relationship between burnout and objective biophysical markers in OMSB residents. Data collection began June 2026 across 13 four-week rotation blocks.

Enrollment: 72 participants (71 active, 1 withdrawn). 100% WHOOP connectivity. 99% baseline completion. 16 residency programs, 3 hospital sites. ~5,000 total data points collected.

2. Cohort Demographics (N = 71)

Characteristic	n (%)		Characteristic	n (%)
Female	44 (62%)		Single	36 (51%)
Male	26 (37%)		Married	32 (45%)
R1	28 (39%)		Has children	16 (23%)
R2	16 (23%)		Exercise 0 days/wk	24 (34%)
R3	14 (20%)		Mean sleep (non-call)	6.2 hours
R4	10 (14%)		Pre-existing psych Hx	5 (7%)

Top programs: Internal Medicine 23, General Surgery 12, Emergency Medicine 6, Pediatrics 5, ENT 5, Psychiatry 4, Anesthesia 3, Orthopedics 3.

3. Baseline Burnout Prevalence

61% of residents met criteria for burnout on at least one CBI subscale (score ≥ 50).

CBI Subscale	Mean	Median	High (≥ 50)	Severe (≥ 75)
Personal	46.7	45.8	26 (37%)	5 (7%)
Work-related	46.2	46.4	31 (44%)	5 (7%)
Patient-related	37.4	37.5	29 (41%)	2 (3%)

Psychological Comorbidity

Instrument	Mean	Moderate or worse	Interpretation
PHQ-9 (Depression)	6.9	19 (27%)	1 in 4 screen positive
GAD-7 (Anxiety)	6.4	20 (28%)	1 in 4 screen positive
ISI (Insomnia)	9.6	12 (17%)	Subthreshold insomnia prevalent
WHO-5 Wellbeing	50.7%	37 (52%) poor	Over half report poor wellbeing

4. Contributing Factors

4.1 Specialty

Program	n	Burnout	Work CBI	HRV (ms)	RHR (bpm)
Anesthesia	3	100%	73.8	46.1	68.9
ENT	5	100%	52.2	39.3	71.8
Emergency Medicine	6	83%	50.6	36.5	69.3
Psychiatry	4	75%	36.6	42.7	74.6
General Surgery	12	67%	52.4	45.0	66.8
Internal Medicine	23	52%	50.3	61.5	64.9
Pediatrics	5	40%	40.0	63.0	61.3
Orthopedics	3	33%	34.5	72.5	60.7

ENT and Emergency Medicine show the lowest HRV in the cohort (39.3 and 36.5 ms), indicating objective autonomic dysregulation consistent with chronic stress.

4.2 Call Patterns (Block 11 Follow-up, N = 58)

Call Type	n	Burnout	Work CBI	PHQ-9	WHOOP Sleep	Recovery
24-hour on-call	25	60%	43.3	7.1	6.0h	57.2%
12-hour shift	18	39%	40.7	4.3	6.3h	60.7%
Home call	2	50%	41.0	13.5	6.9h	62.2%
Other/Mixed	13	46%	33.5	7.2	6.9h	63.0%

24-hour on-call residents have 60% burnout, the shortest objective sleep (6.0h), and lowest recovery scores. 12-hour shift residents show significantly lower burnout (39%) and better recovery.

4.3 Weekly Work Hours

Hours/week	n	Burnout	Work CBI
<40 hours	16	25%	26.8
40-60 hours	31	58%	44.5
60-80 hours	11	64%	47.7

Clear dose-response relationship: burnout more than doubles from <40h (25%) to 60-80h (64%).

4.4 Marital and Parental Status

Factor	n	Burnout	Work CBI	PHQ-9	HRV (ms)
Married	32	66%	48.1	5.8	45.6
Single	36	61%	45.5	8.1	56.6
Has children	16	62%	48.4	5.4	51.2
No children	54	61%	45.5	7.3	53.3

Married residents show higher burnout (66% vs 61%) and significantly lower HRV (45.6 vs 56.6 ms), suggesting work-family conflict as a physiological stressor. Paradoxically, single residents report higher depression (PHQ-9 8.1 vs 5.8), suggesting different distress pathways.

4.5 Gender

Gender	n	Burnout	Work CBI	PHQ-9	GAD-7	HRV	RHR	WHOOOP Sleep
Male	26	69%	44.4	6.2	5.2	55.4	61.8	5.8h
Female	44	57%	47.2	7.2	7.0	51.3	69.0	6.6h

Males have higher burnout prevalence (69% vs 57%) but females report higher depression and anxiety scores. Females sleep nearly 1 hour more per night objectively.

4.6 Exercise

Exercise	n	Burnout	Work CBI	HRV (ms)
0 days/week	24	62%	49.0	47.7
1-2 days/week	29	66%	46.1	53.7
3-4 days/week	16	56%	42.9	59.0

Residents exercising 3-4 days/week show lower burnout (56%) and 24% higher HRV (59.0 vs 47.7 ms) compared to sedentary peers.

4.7 Caffeine

Caffeine/day	n	Burnout	Work CBI	PHQ-9	WHO-5
0	9	33%	41.3	7.1	52%
1-2	48	65%	46.8	6.2	52%
3+	13	69%	47.3	9.1	44%

High caffeine intake (3+/day) is associated with 69% burnout, highest depression (PHQ-9 9.1), and lowest wellbeing (WHO-5 44%).

4.8 Psychiatric History

History	n	Burnout	PHQ-9	GAD-7	WHO-5	ISI
No psych history	66	61%	6.1	5.8	52%	9.2
Psych history	4	75%	19.0	15.0	34%	15.3

Residents with pre-existing psychiatric conditions show dramatically worse scores across all instruments — a vulnerable subgroup requiring proactive support.

4.9 Region of Origin

Region	n	Burnout	PHQ-9	GAD-7
Muscat (local)	31	65%	8.5	7.8
Away from home	39	59%	5.6	5.2

Unexpectedly, Muscat-based residents report higher depression and anxiety than those away from home.

4.10 Major Life Events (Block 11)

Life event	n	Burnout	Work CBI
Yes	15	73%	52.4
No	43	42%	36.0

Residents experiencing a major life event during the block show 73% burnout vs 42% — a 31 percentage-point increase.

5. WHOOP Biophysical Findings

5.1 Cohort Overview (N = 71)

Biometric	Mean	Clinical Interpretation
Recovery Score	59.5%	Yellow zone (suboptimal)

Biometric	Mean	Clinical Interpretation
HRV (rmssd)	52.6 ms	Borderline
Resting Heart Rate	66.3 bpm	Normal
Sleep Duration	6.3 hours	Below recommended 7h
Sleep Efficiency	92.2%	Good
Daily Strain	7.2	Low physical exertion
SpO2	94.5%	Lower boundary of normal

5.2 Burnout Severity Gradient (Work CBI Quartiles)

Quartile	n	CBI Range	HRV (ms)	RHR (bpm)	Recovery	Strain
Q1 (lowest burnout)	17	7-32	58.6	64.9	59.1%	7.1
Q2	17	32-46	55.7	66.6	58.3%	7.4
Q3	17	46-57	48.9	66.7	59.2%	6.9
Q4 (highest burnout)	19	61-93	48.5	67.0	61.5%	7.4

A consistent HRV gradient across burnout quartiles: **17% lower HRV** from Q1 to Q4, with progressive elevation in resting heart rate. This supports burnout as a measurable autonomic phenomenon.

5.3 Self-Reported vs Objective Sleep (N = 70 paired)

Group	Self-Reported	WHOOP-Measured	Discrepancy
All residents	6.2h	6.3h	-0.1h (accurate)
Burnout YES (n=43)	6.1h	6.4h	-0.3h (under-report)
Burnout NO (n=27)	6.4h	6.2h	+0.2h (over-report)

Burned-out residents perceive their sleep as worse than objective measurement shows (-0.3h discrepancy), suggesting altered sleep perception as a burnout feature.

6. Longitudinal Trends: Baseline to Block 11 (N = 58 paired)

Measure	Baseline	Block 11	Change
Work CBI	47.1	40.2	-6.9
Personal CBI	47.0	42.7	-4.2
PHQ-9	6.5	6.5	0.0
GAD-7	6.3	6.1	-0.2
Burnout prevalence	64%	50%	-14%

Who Improved vs Worsened

Trajectory	n (%)	Profile
Improved (CBI down >5)	30 (52%)	60% female, higher exercise rates, PHQ-9 decreased by 0.5
Stable	18 (31%)	—
Worsened (CBI up >5)	10 (17%)	50% male, 40% zero exercise , PHQ-9 increased by +1.6

Sedentary residents were disproportionately represented among those who worsened. Continued monitoring will clarify if exercise is protective or merely a marker of resilience.

7. Data Collection Status

Source	Count	Completion
Baseline assessments	71	99%
Block 11 (follow-up 1)	58/68 eligible	85%
Block 12 (current)	10/70	14% (opened Jul 21)
WHOOP aggregate pulls	2,343	71 active
WHOOP daily rows	2,285	Automated daily
Questionnaire responses	514	CBI + PHQ + GAD + ISI
Weekly check-ins	65	Ongoing
Total data points	~5,000	—

8. Key Conclusions

- Burnout is endemic** — 61% prevalence at baseline, consistent with international literature (35-67% in residents).
- HRV is the strongest biophysical correlate** — burned-out residents show 25% lower HRV (46.7 vs 62.5 ms), with a consistent dose-response gradient across burnout quartiles. This is the study's most significant finding, supporting burnout as measurable autonomic dysfunction.
- 24-hour on-call is the highest-risk work pattern** — 60% burnout, shortest sleep (6.0h), lowest recovery (57.2%). 12-hour shifts show 39% burnout — a 21 percentage-point difference favoring shorter shifts.
- Work hours show a clear dose-response** — burnout more than doubles from <40h/week (25%) to 60-80h/week (64%).
- Specialty matters** — ENT (100%), Anesthesia (100%), and Emergency Medicine (83%) carry the highest burden. Their HRV values (36-46 ms) indicate chronic sympathetic overdrive.
- Marriage increases burnout risk** — married residents show 66% burnout with significantly lower HRV (45.6 vs 56.6 ms), suggesting work-family conflict manifests physiologically.

7. **Exercise is protective** — residents exercising 3-4 days/week show lower burnout (56% vs 62%) and 24% higher HRV. Sedentary residents were over-represented among those who worsened over time.
 8. **Major life events nearly double burnout** — 73% vs 42% during the affected block.
 9. **Sleep perception is altered in burnout** — burned-out residents perceive worse sleep than WHOOP measures, suggesting subjective distortion as a burnout feature.
 10. **Early longitudinal signal is encouraging** — burnout prevalence decreased from 64% to 50% at first follow-up, with 52% of residents showing meaningful improvement.
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Interim report based on data through July 26, 2026. Final analysis will incorporate all 13 rotation blocks, daily WHOOP granularity, multivariate regression, and mediation analysis.

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